

IS4103

Operating Systems

Basic Concepts of Operating Systems

T.Kayalvizhy

Method of Assessments

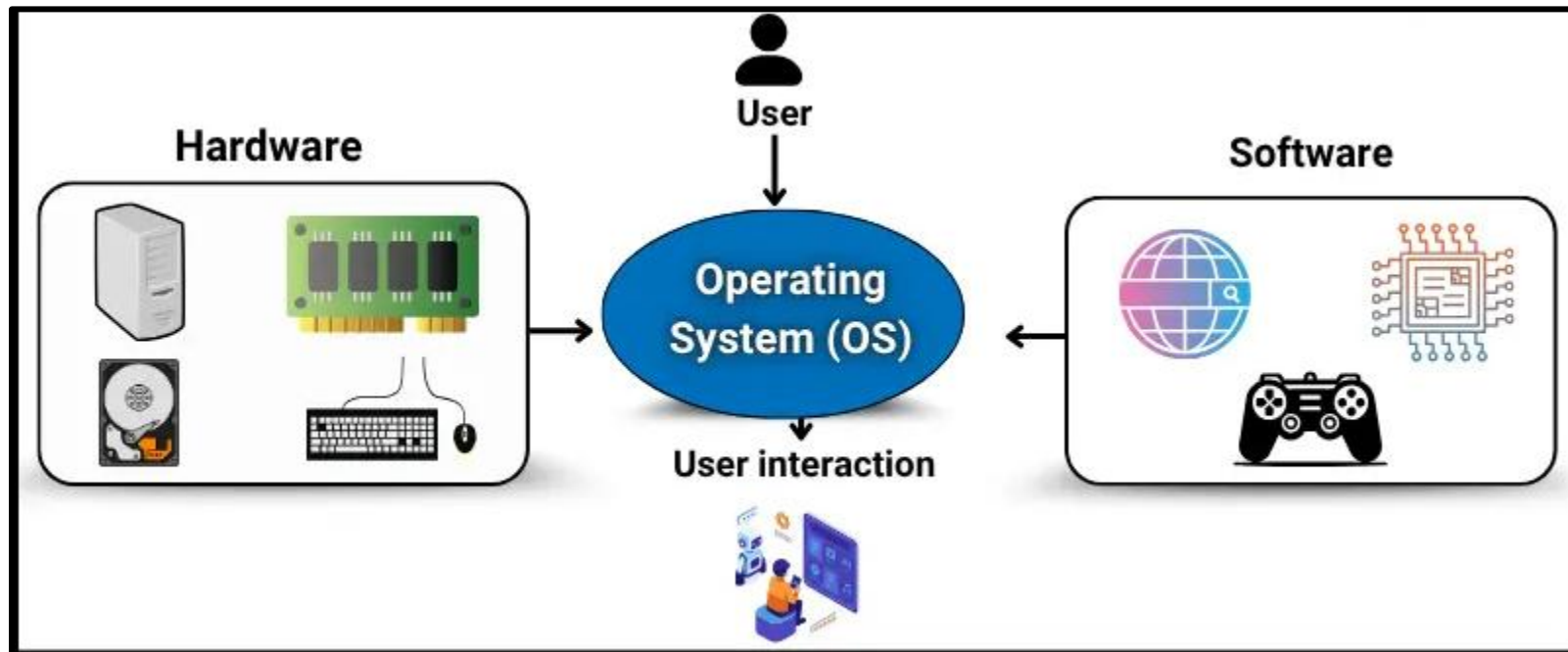
Continuous Assessment (40%)			Final Exam (60%)
Quizzes (%) 20	Assignments (%) 30	Practical Assignments (%) 50	Theory (%) 100

Learning Objectives

Learning Outcome	Description	At the end of Today's Session
ILO1	Describe the basic concepts of operating systems, including development & achievements, functionalities & objectives, structure & components	• What is an Operating System? • Why do we need an Operating System? • Functions of an Operating System

What is an Operating System(OS)?

- An Operating System (OS) is system software that manages computer hardware and software resources and provides common services for computer programs.



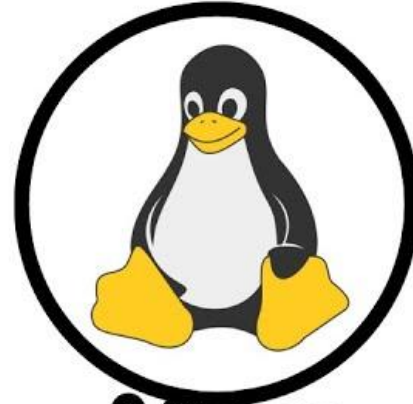
Examples of Operating Systems.



Windows



macOS



Linux



ChromeOS



Android



iOS



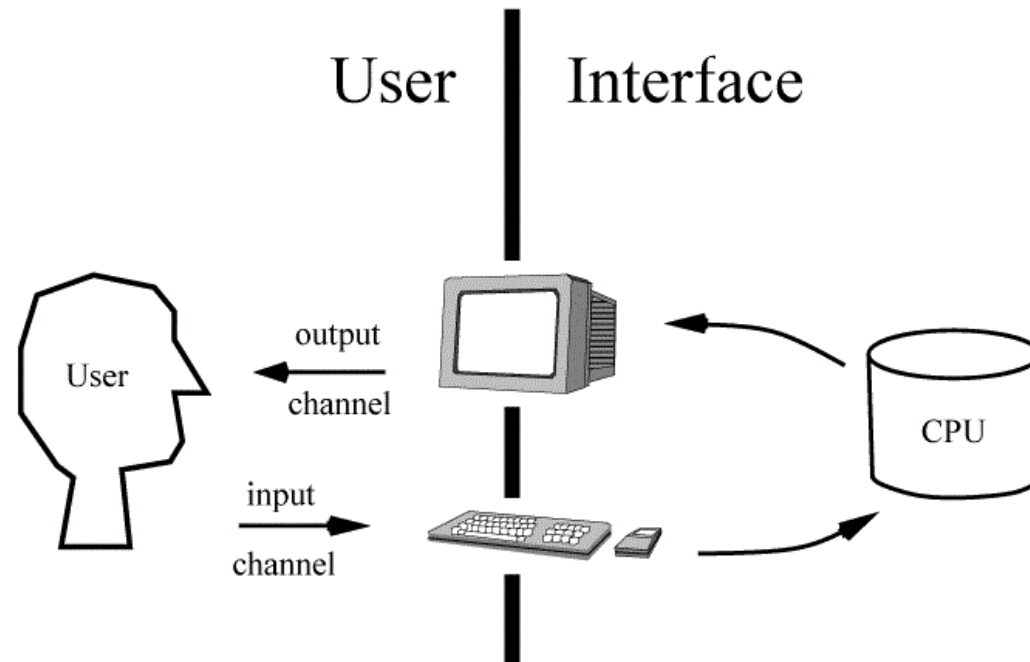
UNIX



BSD

Operating System as an Interface

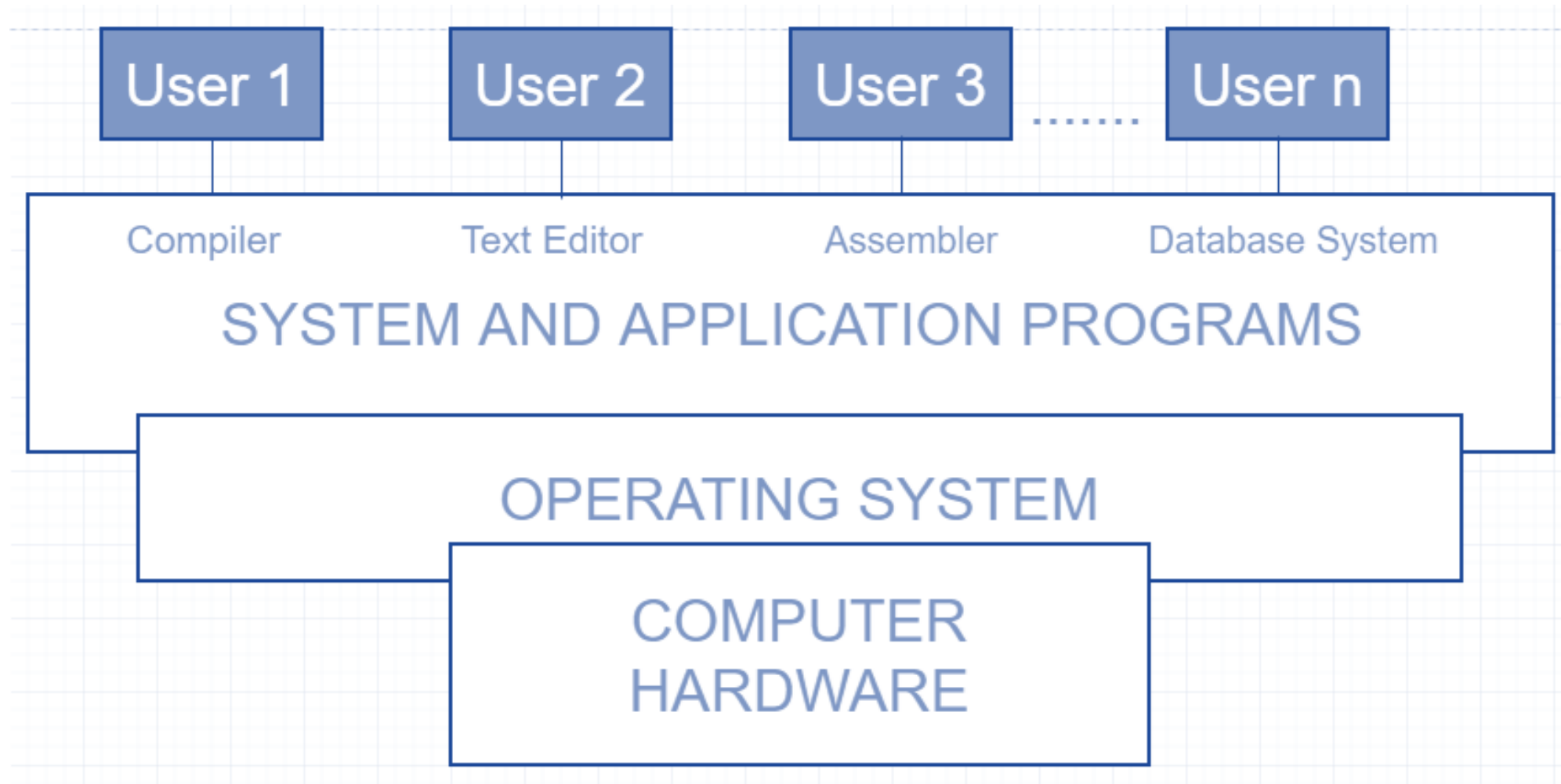
- Users do not interact directly with hardware
- OS acts as an intermediary
- OS translates user commands into hardware actions



Without an Operating System

- If there is no Operating System:
 - Programs must control hardware directly
 - Very complex for users
 - No multitasking
 - No file organization
 - High risk of errors

The process operating system as a User Interface



Why Do We Need an Operating System?

- Main reasons:
 - Convenience for users
 - Efficient use of hardware
 - Resource management
 - Program execution
 - System security

OS for User Convenience

- Provides Graphical User Interface (GUI)
- Provides Command Line Interface (CLI)
- Easy interaction with system
- Reduces technical complexity
 - Examples:
 - Clicking icons
 - Typing commands

What are OS Resources

- Common system resources:
 - Processor (CPU)
 - Main Memory (RAM)
 - Secondary Storage (HDD/SSD)
 - Input Devices (Keyboard, Mouse)
 - Output Devices (Monitor, Printer)

Core Functions of an Operating System

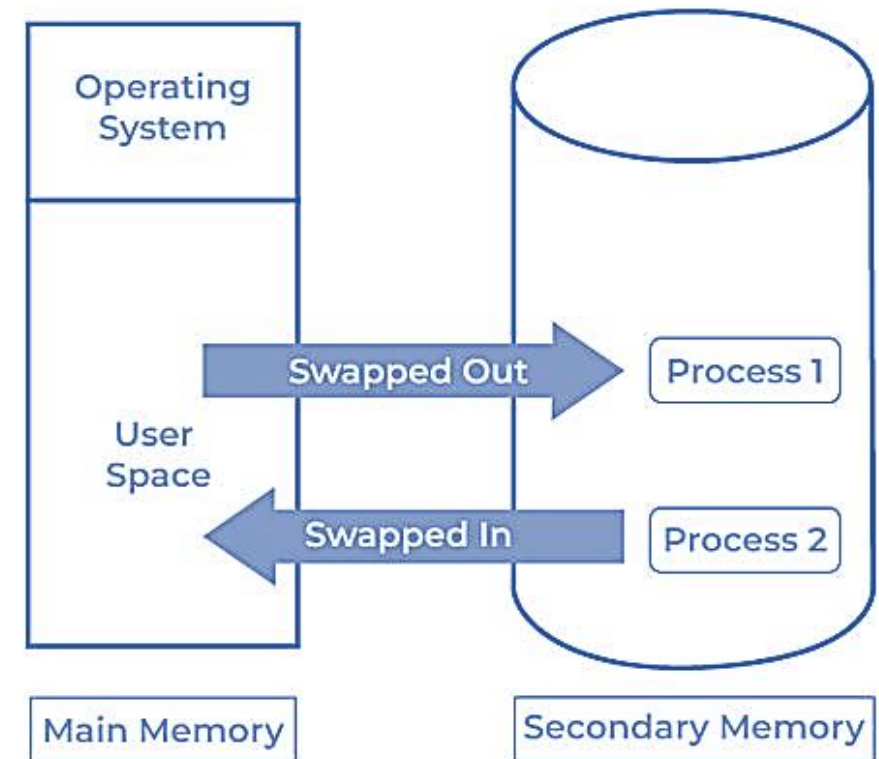
An Operating System (OS) is system software that:

- Controls hardware resources
- Provides services to applications
- Acts as an interface between the **user and the hardware**



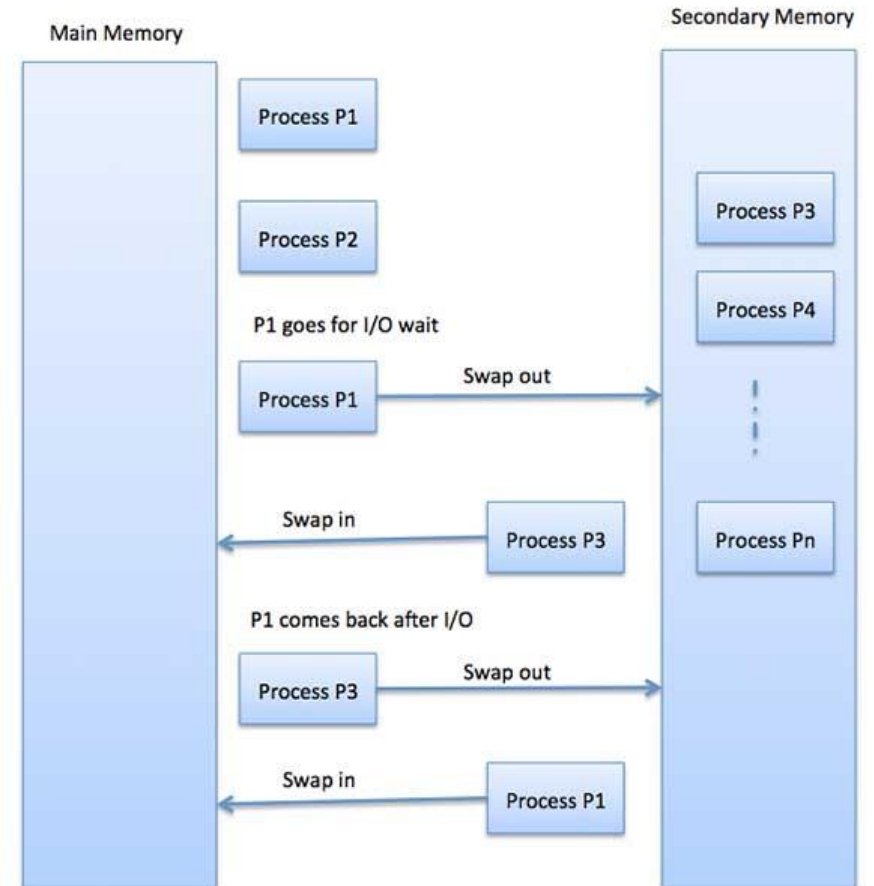
Memory Management

- Memory management is the OS function responsible for:
 - Allocating memory to programs
 - Deallocating memory after use
 - Tracking memory usage

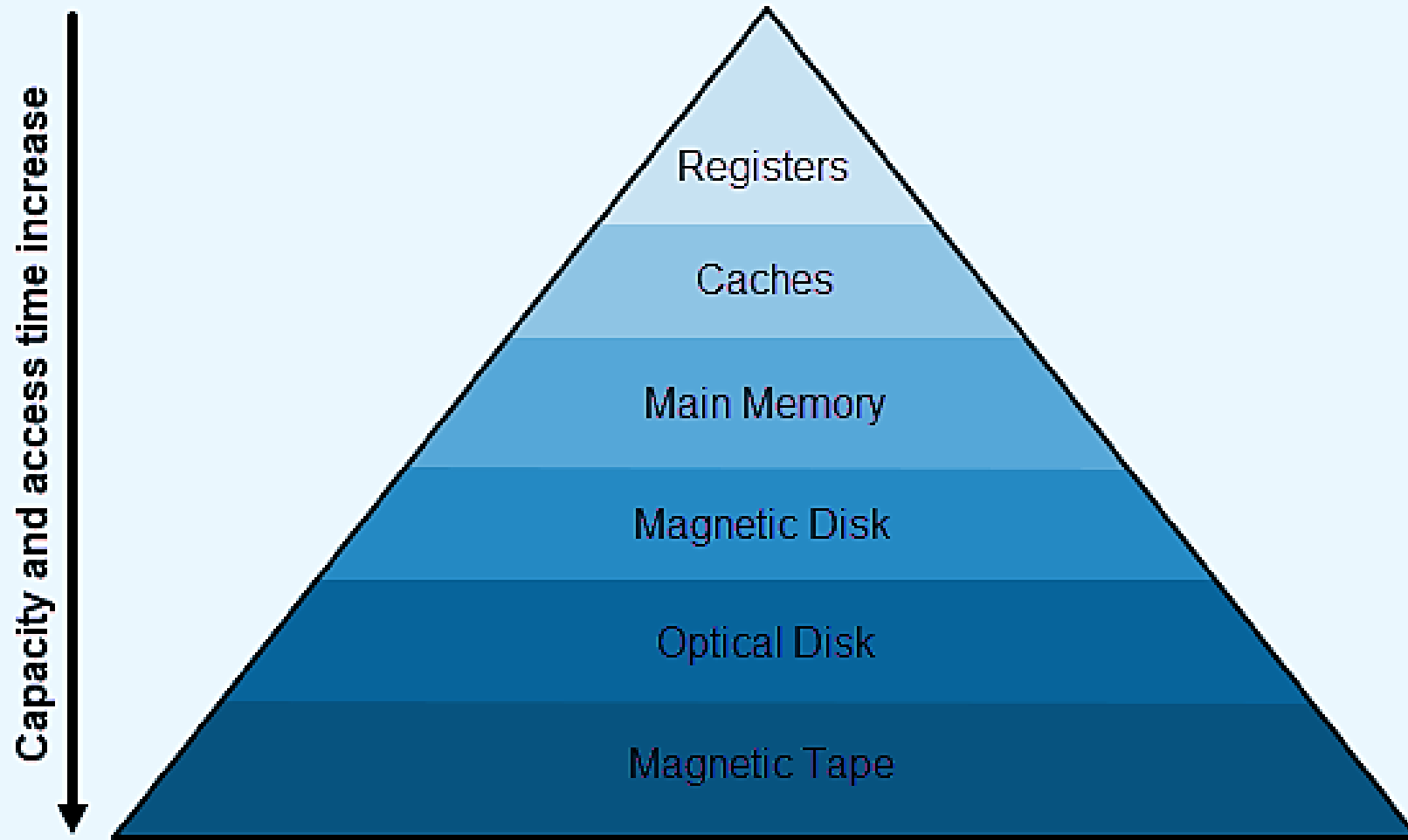


Memory Management –Key function

- Keeps track of which memory locations are free or occupied
- Allocates memory to processes when needed
- Protects one process's memory from another.
- Supports virtual memory (using disk as extension of RAM)

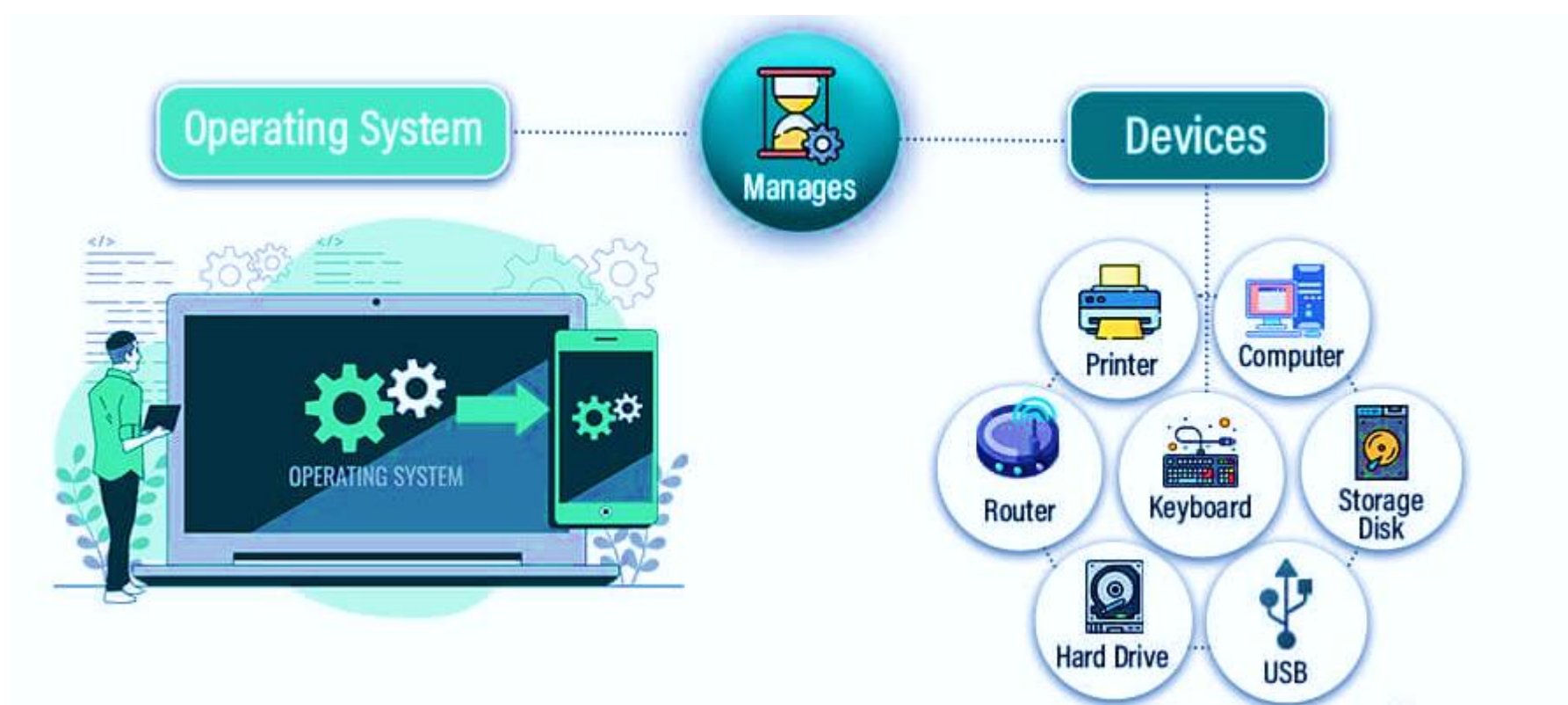


Memory Management in OS



Device Management

- Device management controls all I/O devices such as:



- OS communicates with devices using device drivers

Device Management –Key functions

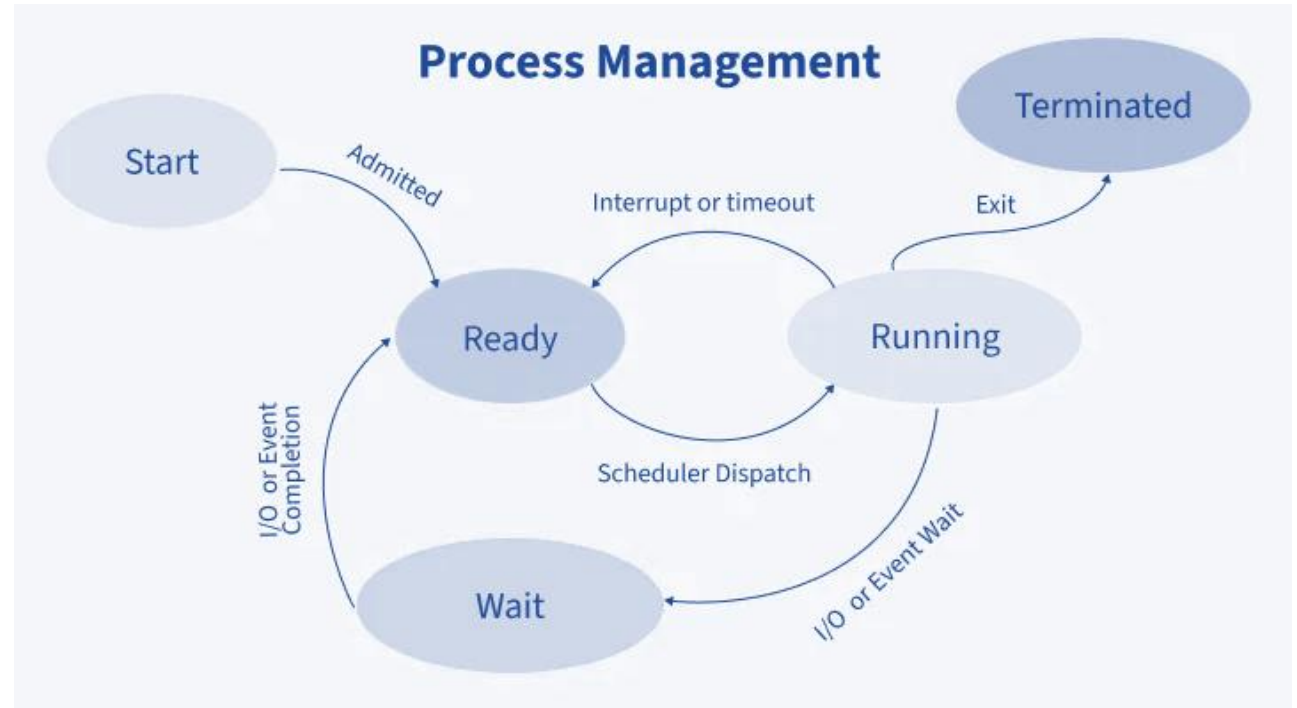
- Allocates devices to processes
- Manages device drivers
- Handles buffering, caching, and spooling
- Resolves device conflicts

Example:



Process Management

- Processor management handles:
 - CPU allocation to processes
 - Process execution and termination
 - Scheduling of tasks



Process Management –Key functions

- Creates and deletes processes
- Schedules processes using algorithms (FCFS, Round Robin, Priority)
- Performs context switching
- Supports multitasking

Example: Switching CPU between browser, media player, and editor.

File Management



File Management

- File management organizes and controls data storage on disks
- Files are logical units of information stored permanently.
- Creation and deletion of files
- Directory management
- File access permissions
- Read, write, and update operations

Example: Organizing files into folders with read/write permissions.

Security Management

- Security protects system resources from:
 - Unauthorized access
 - Malicious attacks
 - Accidental damage
- Essential for multi-user systems
- User authentication (username & password)
- Authorization and access control
- Data protection and encryption
- Auditing and logging

Example: Only admin users can install software.

Error Detection and Handling

- Errors may occur due to:
 - Hardware failures
 - Software bugs
 - User mistakes
- OS must ensure system stability.
- Detects CPU, memory, and I/O errors
- Logs error details
- Takes corrective action (restart, terminate process)
- Prevents system crash

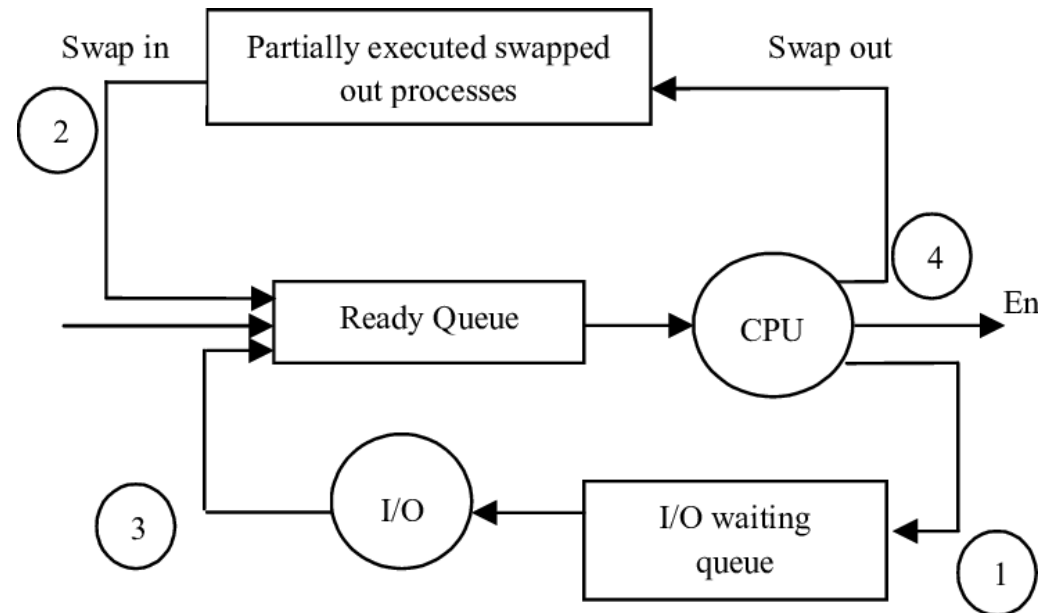
Example: Handling disk read/write errors safely.

Coordination Between User, Software, and Hardware

- OS as a Coordinator
OS acts as an intermediary between:
 - User
 - Application software
 - Hardware components
- Users do not interact directly with hardware.

Job Accounting/ Scheduling

- Job accounting tracks:
 - Resource usage by users and processes
 - CPU time, memory usage, I/O operations
- Important for system monitoring and billing



Job Accounting – Key Functions

- Maintains logs of resource usage
- Helps in performance analysis
- Supports fair resource allocation
- Useful in multi-user and server systems

Example: Universities tracking student server usage.

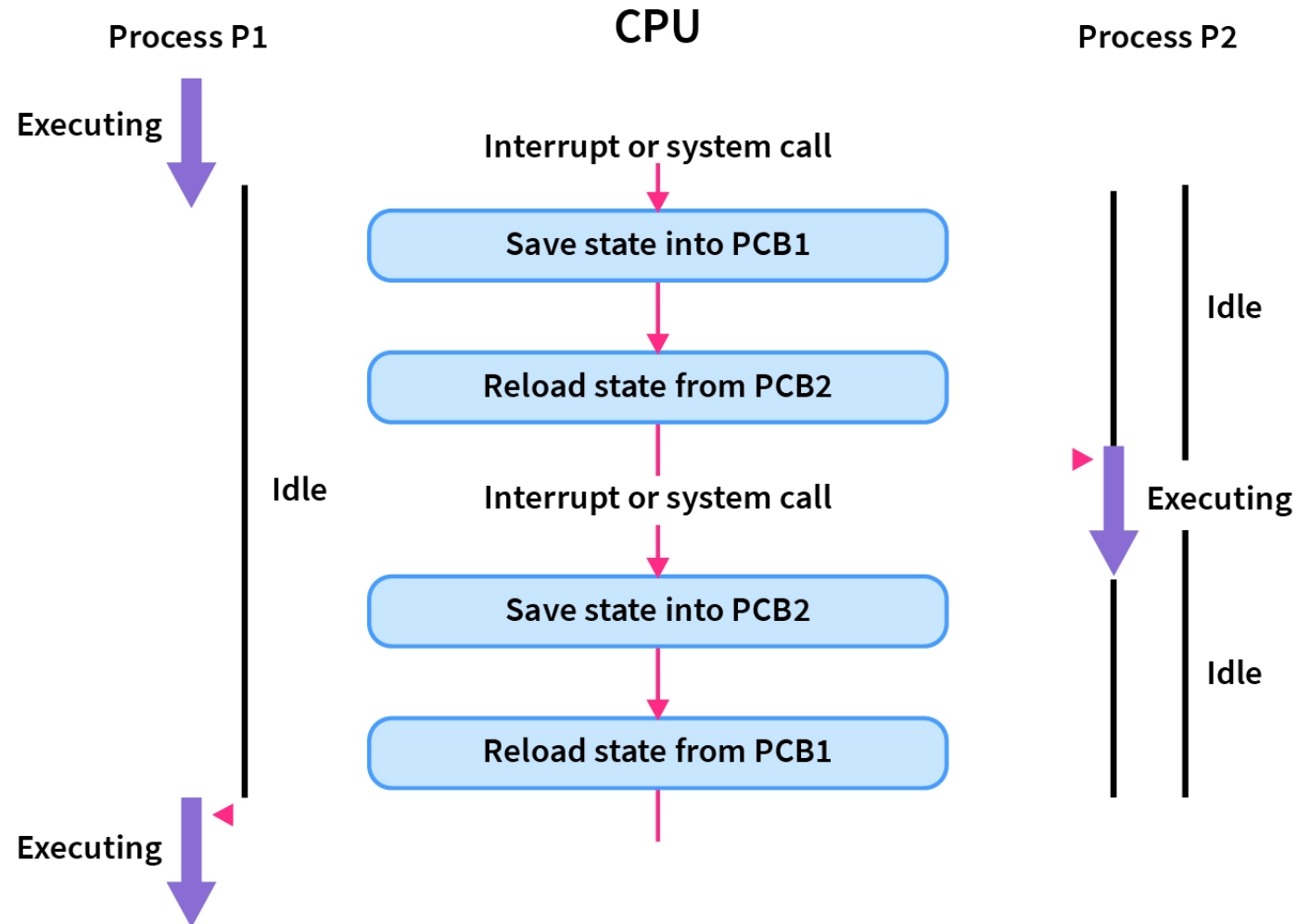
Context Switching

- Context switching is the process of switching the CPU from one process, task, or thread to another.
- Example:

Switching between meetings and tasks. You're working on an important task when you're called into an impromptu meeting.



Context Switching



Key Takeaways

- Memory management ensures efficient RAM usage
- Processor management controls CPU execution
- Device management handles I/O devices
- File management organizes storage
- Security protects system resources
- Error detection maintains stability
- OS coordinates user, software, and hardware
- Job accounting monitors resource usage

Thank you

Any Questions?